



USER MANUAL v1.0

GRIMs-NEMO Fully Coupled Model v1.0 Manual

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1. System Overview

The GRIMs-NEMO system is a fully coupled atmosphere-ocean-sea-ice model. GRIMs v3.2 provides the atmospheric component, NEMO v4.2.2 provides the ocean component, and the Sea Ice Modelling Integrated Initiative (SI3) provides the sea-ice component. OASIS3-MCT exchanges coupling fields between GRIMs-GMP and NEMO/SI3.

Configuration	NEMO v4.2.2	GRIMs v3.2
Horizontal resolution	eORCA025	T126
Vertical levels	75	28
Input data	NEMOVAR restart file	ERA5 reanalysis and OISST
Time step	600 sec	300 sec
Fields sent to the other component	Sea surface temperature, sea-ice fraction, and snow depth	Wind stress; solar heat flux; rain; snow; evaporation; sensible heat flux; latent heat flux; and longwave flux
Coupling interval	1 h	1 h

2. Installation

2.1 Required Components and Installation Order

Install the coupler, ocean model, and atmospheric model on the target server. Because both NEMO and GRIMs are compiled against the OASIS3-MCT libraries, OASIS3-MCT must be installed first. NEMO and GRIMs may then be installed in either order.

1. Install OASIS3-MCT.
2. Install NEMO.
3. Install GRIMs-GMP.

The source environment used the following software stack. Verify compatible versions before installation:

1. Fortran, C, and C++ compilers: Intel compiler toolchain
2. NetCDF: intel/netcdf-4.x
3. MPI: intel/intelmpi
4. HDF5: gcc/hdf5-1.10.5

Extract the three source archives:

```
tar -xvf nemo_cpl.tar
tar -xvf gmp_cpl.tar
tar -xvf oasis3-mct.tar
```

2.2 Installing OASIS3-MCT

1. Download or unpack the OASIS3-MCT source code.
2. Move to the make-file directory and open the site configuration file.

```
cd util/make_dir
vi make.roses
```

In `make.inc`, specify the absolute path to the file that will be used as the make configuration. A supplied make file for a similar computing environment may be copied and adapted.

```
#
# System dependent settings
#
# $Id: make.inc 1123 2006-09-06 12:14:56Z valcke $
#
##### User configurable options #####
#
# Note: The absolute path name must be indicated.
#
# Note: Choose one of these includes files and modify it according to your
#       local settings. Replace the currently active file with your own.
#
include $(HOME)/NEMO/oasis3-mct/util/make_dir/make.roses
#
### End User configurable options ###
```

Figure 1. `make.inc` pointing to the absolute path of the selected OASIS3-MCT make file.

Edit the selected make file after checking the compiler, MPI, NetCDF, and other library paths available on the server. Then compile OASIS3-MCT:

```
make -f TopMakefileOasis3
```

```
# CHAN : communication technique used in OASIS3 (MPI1/MPI2)
CHAN = MPI1
#
# Paths for libraries, object files and binaries
#
# COUPLE : path for oasis3-mct main directory (defined in .bash_profile)
COUPLE = /rhome/soo/NEMO/oasis3-mct
#
# ARCHDIR : directory created when compiling
ARCHDIR := $(COUPLE)/INSTALL_OASIS
#
# MPI library # 27/10/2020: /data/softs/intel/impi/2018.1.163/ (module load mpi/intelmpi/2018.1.163-ddt)
MPIRUN = mpiexec
#
# NETCDF library
#NETCDF_LIB_DIR=
NETCDF_INC_DIR=/usr/local/netcdf/4.13.0/include # 27/10/2020: /data/softs/local_intel18/netcdf/4.4.4_phdf5_1.10.4/include (module load lib/netcdf-fortran/4.4.4_phdf5_1.10.4)
NETCDF_INCLUDE = ${NETCDF_INC_DIR}
NETCDF_LIBRARY = `nf-config --flibs` # 27/10/2020: -L/data/softs/local_intel18/netcdf/4.4.4_phdf5_1.10.4/lib -lnetcdff -L/data/softs/local_intel18/phdf5/1.10.4_impi/lib -L/data/softs/local_intel18/netcdf/4.4.4_phdf5_1.10.4/lib -Wl,-rpath,/data/softs/local_intel18/phdf5/1.10.4_impi/lib -Wl,-rpath,/data/softs/local_intel18/netcdf/4.4.4_phdf5_1.10.4/lib -lhdf5_hl -lhdf5 -lz -lcurl -lnetcdff -lnetcdf
#
# Compiling and other commands
MAKE = gmake
F90 = mpiifort # 27/10/2020: /data/softs/intel/impi/2018.1.163/bin64/mpiifort -va chercher /data/softs/intel/compilers_and_libraries_2018.1.163/linux/bin/intel64/fort (module load compiler/intel/18.0.1.163)
F = $(F90)
f90 = $(F90)
f = $(F90)
CC = mpicc
LD = mpiifort
AR = ar
ARFLAGS = -ruv
```

Figure 2. Example OASIS3-MCT make-file settings for the source server. (*make.roses*)

After a successful build, confirm that module files (`\*.mod`) have been created under `oasis3-mct/INSTALL\_OASIS/include`.

```

-rw-rw-r-- 1 soo soo 13310 2023-01-30 21:28 remap_write.mod
-rw-rw-r-- 1 soo soo 52153 2023-01-30 21:28 remap_vars.mod
-rw-rw-r-- 1 soo soo 37437 2023-01-30 21:28 remap_locc_weight.mod
-rw-rw-r-- 1 soo soo 38868 2023-01-30 21:28 remap_distance_gaussian_weight.mod
-rw-rw-r-- 1 soo soo 10515 2023-01-30 21:29 mod_oasis_load_balancing.mod
-rw-rw-r-- 1 soo soo 2968 2023-01-30 21:29 mod_oasis_kinds.mod
-rw-rw-r-- 1 soo soo 268 2023-01-30 21:29 mod_oasis_ioshr.mod
-rw-rw-r-- 1 soo soo 49250 2023-01-30 21:29 mod_oasis_io.mod
-rw-rw-r-- 1 soo soo 71174 2023-01-30 21:29 mod_oasis_grid.mod
-rw-rw-r-- 1 soo soo 21697 2023-01-30 21:29 mod_oasis_getput_interface.mod
-rw-rw-r-- 1 soo soo 50527 2023-01-30 21:29 mod_oasis_data.mod
-rw-rw-r-- 1 soo soo 72995 2023-01-30 21:29 mod_oasis_coupler.mod
-rw-rw-r-- 1 soo soo 43454 2023-01-30 21:29 mod_oasis_auxiliary_routines.mod
-rw-rw-r-- 1 soo soo 65390 2023-01-30 21:29 mod_oasis_advance.mod
-rw-rw-r-- 1 soo soo 46091 2023-01-30 21:29 mod_oasis.mod
-rw-rw-r-- 1 soo soo 45652 2023-01-30 21:29 mod_prism.mod
-rw-rw-r-- 1 soo soo 17173 2023-01-30 21:29 mod_oasis_var.mod
-rw-rw-r-- 1 soo soo 3713 2023-01-30 21:29 mod_oasis_timer.mod
-rw-rw-r-- 1 soo soo 5548 2023-01-30 21:29 mod_oasis_sys.mod
-rw-rw-r-- 1 soo soo 14887 2023-01-30 21:29 mod_oasis_string.mod
-rw-rw-r-- 1 soo soo 11163 2023-01-30 21:29 mod_oasis_reprosum.mod
-rw-rw-r-- 1 soo soo 39455 2023-01-30 21:29 mod_oasis_part.mod
-rw-rw-r-- 1 soo soo 15680 2023-01-30 21:29 mod_oasis_parameters.mod
-rw-rw-r-- 1 soo soo 11105 2023-01-30 21:29 mod_oasis_namcouple.mod
-rw-rw-r-- 1 soo soo 52738 2023-01-30 21:29 mod_oasis_mpi.mod
-rw-rw-r-- 1 soo soo 38638 2023-01-30 21:29 mod_oasis_method.mod
-rw-rw-r-- 1 soo soo 2392 2023-01-30 21:29 mod_oasis_mem.mod
-rw-rw-r-- 1 soo soo 45549 2023-01-30 21:29 mod_oasis_map.mod
[roses]:soo:/rhome/soo/NEMO/oasis3-mct/INSTALL_OASIS/include>

```

Figure 3. OASIS3-MCT module files created under `INSTALL\_OASIS/include`.

To remove the existing OASIS3-MCT build and compile again, use:

```
make realclean -f TopMakefileOasis3
```

2.3 Installing NEMO

2.3.1 Configure the Architecture File

Move to the NEMO `arch/` directory. Select an `arch-\*.fcm` file that most closely matches the target system, copy it under a site-specific name, and edit it for the local environment.

```
cd arch/
cp /rhome/soo/NEMO/nemo-4.2.0/arch/arch-linux_ifort.fcm ./arch-roses_ifort.fcm
```

Enter absolute paths for each required library. For an unused library, retain an appropriate dummy path as in the source configuration. NetCDF and OASIS are mandatory for the coupled model. Verify that the compiler and linker commands are consistent with the selected environment.

```

%NCDF_HOME      /usr/local/netcdf/461_intel21
%HDF5_HOME      /usr/local/hdf5/1.10.5
%XIOS_HOME      /not/defined
%OASIS_HOME      /rhome/soo/NEMO/oasis3-mct/INSTALL_OASIS

%NCDF_INC        -I%NCDF_HOME/include
%NCDF_LIB        -L%NCDF_HOME/lib -lnetcdff -lnetcdf -L%HDF5_HOME/lib -lhdf5_hl -lhdf5 -lhdf5
%XIOS_INC        -I%XIOS_HOME/inc
%XIOS_LIB        -L%XIOS_HOME/lib -lxios -lstdc++
%OASIS_INC       -I%OASIS_HOME/build-static/lib/mct -I%OASIS_HOME/build-static/lib/psmile.MPI1
%OASIS_LIB       -L%OASIS_HOME/lib -lpsmile.MPI1 -lmct -lmpeu -lscrip

%CPP             cpp
%FC             mpiifort
##%FCFLAGS       -i4 -r8 -O0 -fp-model precise -fno-alias -mcmodel=medium -assume byterecl -g -traceback -check all
%FCFLAGS         -i4 -r8 -O3 -g -traceback -fp-model precise -fno-alias -mcmodel=medium -assume byterecl
%FFLAGS         %FCFLAGS
%LD             mpiifort
%LD_FLAGS        -P -traditional
%FPPFLAGS        -P -traditional
%AR             ar
%ARFLAGS         rs
%MK             gmake
%USER_INC        %XIOS_INC %OASIS_INC %NCDF_INC
%USER_LIB        %XIOS_LIB %OASIS_LIB %NCDF_LIB

%CC             cc
%CFLAGS         -O0

```

Figure 4. Example `arch-roses\_ifort.fcm` library paths, compiler commands, and flags.

2.3.2 Select NEMO CPP Keys

NEMO contains ocean, sea-ice, biogeochemical, and other components; compile only the code required by the selected configuration. Template configurations and test cases are available under `cfgs/` and `test/`. The coupled-model example uses `test\_CPL\_OASIS`.

```

cd test/test_CPL_OASIS
vi cpp_test_CPL_OASIS.fcm

```

1. **key_oasis3**: required for OASIS3-MCT coupling.
2. **key_si3**: required when the SI3 sea-ice model is used.

```
bld::tool::fppkeys key_oasis3 key_qco key_si3
```

Figure 5. Required CPP keys for OASIS3-MCT and SI3.

2.3.3 Compile NEMO

```
./makenemo -m roses_ifort -a test_CPL_OASIS -n <run_directory_name> [-j
<number_of_threads>]
```

Replace `roses\_ifort` with the suffix used in the selected `arch-\*.fcm` file. If `` is set to `test\_CPL\_OASIS`, NEMO compiles in the existing test-case directory instead of creating a new directory under `test/`.

After compilation, confirm the successful-build message and verify that `nemo.exe` exists under `/BLD/bin/`.


```

fcm_internal load:F nemo nemo.o nemo.exe
ar: creating /rhome/soo/NEMO/couple/couple_ice/nemo-4.2.2/test
mpifort -o nemo.exe /rhome/soo/NEMO/couple/couple_ice/nemo-4.
t_CPL_OASIS/BLD/lib -l__fcm__nemo -L/not/defiled/lib -lstdc+
netcdf/461_intel21/lib -lnetcdff -lnetcdf -L/usr/local/hdf5/1.
->Make: 909 seconds
->TOTAL: 909 seconds
Build command finished on Wed Jul 10 18:13:13 2024.
/rhome/soo/NEMO/couple/couple_ice/nemo-4.2.2/tests

```

Figure 6. Successful NEMO build message.

2.4 Installing GRIMs-GMP

A GRIMs-GMP build requires the `libs\_gmp`, `srcs\_gmp`, and `runs\_gmp` directories. Extracting `gmp\_cpl.tar` also provides `run\_gmp\_restart`, which contains code used to restart the coupled model.

2.4.1 Compile `libs\_gmp`

```

cd libs_gmp
vi configure-lib

```

Edit `configure-lib` for the target system. The source example uses settings for the Synoptic Meteorology Laboratory `roses` server.

```

#!/bin/sh
#set -ax
#
#-----
#
# grims lib-script configuration
#
#   Program history :
#
#   12-19-2000   song-you hong & masao kanamitsu   initial development
#   03-19-2011   song-you hong & nml group         grims setup
#
#-----
#
# Change the following variable
#
# MACHINE=[sgi/origin/ibmsp/sun/dec/hp/cray/t90/t3e/linux_pgi/linux_intel/x1e]
# MARCH=[single,thread,mpi,hybrid,mac]
# INSTITUTION=[sml,nml,kisti,kma,kesti,ncar,apcc,snu,noaa,kiaps,kiapsrt,haenam]
#
MACHINE=linux_intel
MARCH=mpi
INSTITUTION=nml
USERNAME=baram
#
#
# If special options for compilation is needed, change below
#
OPTIONS_FILE=./opt_libs/options-$MACHINE-$MARCH
if [ -s $OPTIONS_FILE-$INSTITUTION ]; then
    cp $OPTIONS_FILE-$INSTITUTION $OPTIONS_FILE
fi
#
#=== End of user edit section =====

```

Figure 7. Example `configure-lib` system settings for `libs\_gmp`.


```
./configure-lib
make
```

A successful `./configure-lib` run creates a Makefile. After `make` completes, verify that executable files (*.x) have been generated under `etc/`.

```
-rwxr-xr-x 1 soo soo 120296 2021-01-28 19:50 ieee2grb_sst.x*
-rwxr-xr-x 1 soo soo 10204 2021-01-28 19:50 incdte.x*
-rwxr-xr-x 1 soo soo 10825 2021-01-28 19:50 inchour.x*
-rwxr-xr-x 1 soo soo 120331 2021-01-28 19:50 ieee2grb_sm.x*
-rwxr-xr-x 1 soo soo 115221 2021-01-28 19:50 ieee2grb.x*
-rwxr-xr-x 1 soo soo 42299 2021-01-28 19:50 prmap.x*
-rwxr-xr-x 1 soo soo 168113 2021-01-28 19:50 sigphys.x*
-rwxr-xr-x 1 soo soo 127905 2021-01-28 19:50 scnvrt.x*
-rwxr-xr-x 1 soo soo 38767 2021-01-28 19:50 rsigdiff.x*
-rwxr-xr-x 1 soo soo 32208 2021-01-28 19:50 interp_subr.o*
-rwxr-xr-x 1 soo soo 238533 2021-01-28 19:50 gribinterp.x*
-rwxr-xr-x 1 soo soo 133972 2021-01-28 19:50 scnvrt_gfs.x*
-rwxr-xr-x 1 soo soo 194847 2021-01-28 19:50 scnvrt_gsi.x*
-rwxr-xr-x 1 soo soo 18338 2021-01-28 19:50 sigdif.x*
-rwxr-xr-x 1 soo soo 18338 2021-01-28 19:50 sigadd.x*
-rwxr-xr-x 1 soo soo 24871 2021-01-28 19:50 force_grib_date_mon*
-rwxr-xr-x 1 soo soo 166703 2021-01-28 19:50 wgrib*
-rwxr-xr-x 1 soo soo 29294 2021-01-28 19:50 maketim*
-rwxr-xr-x 1 soo soo 24785 2021-01-28 19:50 sdir*
-rwxr-xr-x 1 soo soo 24875 2021-01-28 19:50 force_grib_date_mon_fhr*
-rwxr-xr-x 1 soo soo 27503 2021-01-28 19:50 mpiset.x*
drwxrwxr-x 3 soo soo 4096 2021-01-28 19:50 utils/
-rwxr-xr-x 1 soo soo 22388 2021-01-28 19:50 rmpiset.x*
[roses]:soo:/rhome/soo/GRIMs/gmp/libs_gmp/etc>
```

Figure 8. Example executables generated in `libs\_gmp/etc`.

Recompilation rule: Run `make` again after modifying source code. If the configure settings change, run `make clean` before recompiling.

2.4.2 Compile `srcs\_gmp`

```
cd srcs_gmp
vi configure-src
```

Variable	Description	Setting
LIBS_DIR	Path to the library directory	Path to the compiled `libs_gmp` directory
OASIS_LIB	Absolute path to the OASIS3-MCT library	Path to the installed OASIS3-MCT library
OASIS_INC	Absolute path to the OASIS3-MCT include directory	Path to the installed OASIS3-MCT include directory
NETCDF_LIB	Absolute path to the NetCDF library	Path to the server's NetCDF library
MODEL	Model type based on spherical harmonics	Use `sph` for the global model
VERSION	Physics package option	Use `V3.2`
MARCH	CPU/build type	Must match the value used in `configure-lib`

```
#!/bin/sh
#set -ax
#-----
#
# grims src-script configuration
#
# Program history :
#
# 12-19-2000 song-you hong & masao kanamitsu initial development
# 03-19-2011 song-you hong & nml group members grims setup
#-----
#
# LIBS_DIR .... Model library, utility executables, scripts
#               and constants directory (where configure-lib
#               and make are executed). This directory should
#               already exist.
#
LIBS_DIR=/rhome/soo/NEMO/couple/couple_ice/gmp_v3.2/libs_gmp
#soojin_couple
OASIS_LIB=/rhome/soo/NEMO/oasis3-mct/INSTALL_OASIS/lib
OASIS_INC=/rhome/soo/NEMO/oasis3-mct/INSTALL_OASIS/build-static/lib/psmile.MPI1
NETCDF_LIB=/usr/local/netcdf/461_intel21/lib
#####
#
# MODEL .... sph, dfs, nim, mul, rmp, smp, ssi, gsi, gop, rop, chm
#
# sph : spherical harmonics gmp
# dfs : double fourier spectral gmp
# nim : non-hydrostatic icosahedral model (noaa/esrl)
# mul : aqua-planet testbed - neale and hoskins (2001, asl)
# rmp : regional model program
# gop : global ocean model program (mom3)
# gom : global ocean model program (mom4p1)
# rop : regional ocean model program
# smp : single column model program
# ssi : spectral statistical interpolation data assimilation
# gsi : grid point statistical interpolation data assimilation
# chm : global model program with aerosol chemistry module
#
MODEL=sph
#
# define model version
#   = /v1.0/v2.0/v3.0/v4.0/v5.0 (see doc/readme files for sub-versions)
#
VERSION=v3.2
#
# MARCH .... select from single/mpi/thread/hybrid
#
MARCH=mpi
#
# define smp data .... select from arms18/arms38/gate19/toga40/trmm37
#
```

Figure 9. Example `configure-src` settings for GRIMs-GMP.

define.h: Running the configure step creates or resets `define.h`. Physical-process schemes are selected in this file. If `define.h` is edited, run `make` again. Because a subsequent configure step reinitializes the file, preserve any required changes.

```
./configure-src
make
```

After `./configure-src`, verify that both `Makefile` and `define.h` exist. After compilation, verify that executable files (`\*.x`) have been generated under `bin/`.

```
drwxrwxr-x 3 soo soo 4096 2020-07-16 14:24 makefiles/
drwxrwxr-x 3 soo soo 4096 2020-07-16 14:24 doc/
drwxrwxr-x 4 soo soo 4096 2020-07-16 14:24 def/
lrwxrwxrwx 1 soo soo 19 2020-07-16 14:24 configure-src -> ./env/configure-src*
drwxrwxr-x 22 soo soo 4096 2020-07-16 14:45 src/
drwxrwxr-x 2 soo soo 6 2021-02-01 15:51 bin/
drwxrwxr-x 3 soo soo 37 2021-02-01 15:51 env/
-rw-rw-r-- 1 soo soo 20 2021-02-01 15:51 machine.h
-rwxrwxr-x 1 soo soo 1375 2021-02-01 15:51 options*
-rw-rw-r-- 1 soo soo 185 2021-02-01 15:51 cvar.env
-rw-rw-r-- 1 soo soo 65 2021-02-01 15:51 define_mpi.h
-rw-rw-r-- 1 soo soo 16843 2021-02-01 15:51 define.h
-rw-rw-r-- 1 soo soo 1245 2021-02-01 15:51 Makefile
```

Figure 10. `Makefile` and `define.h` created by `./configure-src`.

```
15:54 share.a
15:54 common.a
15:54 common_par.a
15:54 mpi.a
15:54 sfcl.a
15:54 sfcl_par.a
15:55 chgr.x*
15:55 chgdates.x*
15:55 co2.x*
15:55 mtn.x*
15:55 pgb.x*
15:57 fcst.x*
15:57 sgb.x*
15:57 fgb.x*
15:57 sfc0.x*
15:57 change_resolution_pressure.x*
```

Figure 11. Example executables generated in `srcs\_gmp/bin`.

Verification required: The source states that `make clean` deletes built files but retains `define.h`, and gives `make dist\_celan` as the command to delete `define.h` as well. Confirm the exact clean target in the active Makefile before use.

2.4.3 Configure and Compile `runs\_gmp`

```
cd runs_gmp
vi configure-run
```

Variable	Description	Setting
SRCS_DIR	Path to the source-code directory	Path to the compiled `srcs_gmp` directory
OUTPUT_DIR	Output-directory path	Specify a path if a fixed output location is required
POST_DIR	Postprocessing-directory path	Specify a path if a fixed postprocessing location is required
WAVE	Spectral wave number controlling horizontal resolution	62: 2.5° x 2.5° 126: 1° x 1° 254: 0.5° x 0.5° 510: 0.25° x 0.25°
MODEL_LAYER	Number of vertical levels	Select the required value
POST_IGRD, POST_JGRD	Grid dimensions of output files	Set for the chosen resolution using the comments in the configure file
POST_LAYER	Number of pressure-output levels	Normally 26

Variable	Description	Setting
ILONF, ILATG, IJCAP, ILEVS	Horizontal grid, wave number, and vertical levels of P2SIG `sfc.anl.*.ieee` input	`ILONF`, `ILATG`: SFC file `IJCAP`, `ILEVS`: SIG file
NCPUS	Number of threads	Fixed at 1 in the source setup
NPES	Number of CPUs assigned to GRIMs	The `roses` server has 36 CPUs per node
NCOL	Number of zonal subdivisions in the horizontal domain	Use 4 for at least 126 CPUs; otherwise use 2
NCPU_PER_NODE	Number of CPUs per node	36 on the `roses` server

Because the coupled system is launched from `gmp\_runs`, link the NEMO executable and namelists through the GRIMs run scripts. Enter the NEMO installation path at the indicated location in `runscr/fcst.in`.

```
vi runscr/fcst.in
```

```
#####
#soojin_couple
#for nemo run
@FTASS@ /rhome/soo/NEMO/couple/couple_ice/nemo-4.2.2/tests/test_CPL_OASIS/BLD/bin/nemo.exe ./nemo
@FTASS@ /rhome/soo/NEMO/couple/couple_ice/nemo-4.2.2/tests/test_CPL_OASIS/EXP00/namelist_cfg .
@FTASS@ /rhome/soo/NEMO/couple/couple_ice/nemo-4.2.2/tests/test_CPL_OASIS/EXP00/namelist_ref .
@FTASS@ /rhome/soo/NEMO/couple/couple_ice/nemo-4.2.2/tests/test_CPL_OASIS/EXP00/namelist_ice_cfg .
@FTASS@ /rhome/soo/NEMO/couple/couple_ice/nemo-4.2.2/tests/test_CPL_OASIS/EXP00/namelist_ice_ref .
#####
```

Figure 12. NEMO executable and namelist links in `runscr/fcst.in`.

```
cd /rhome/soo/GRIMs/gmp/runs_gmp/
./configure-run gmp
```

The configure step compiles the run layer and creates the `gmp` script/executable in `runs\_gmp`.

```
configure-run -> ./env/configure-run*
test_case/
typhoon_case/
log
expscr/
env/
runscr/
options*
cvar.env
gmp*
namelist.gmp
```

Figure 13. Example `runs\_gmp` directory after `./configure-run gmp`.

3. Running the Coupled Model

3.1 Required Initial and Boundary Data

Both atmospheric and oceanic input datasets are required at the coupled-model start time. The source workflow uses the following inputs.

Component	Required input	Use or preprocessing
GRIMs	Initial surface file	R2 surface data used by P2SIG
GRIMs	Pressure-level file	GRIB input used by P2SIG
GRIMs	Initial SST file	GRIB format
NEMO	Initial ocean temperature and salinity	Interpolate to the NEMO grid using `1.input_interp.csh` and `2.interp_to_orca.csh` under `/rhome/soo/NEMO/input/make_input/make_ocean_init/`
NEMO	Surface boundary conditions	Process ERA5 using `3.ERA5_interp.csh` and `4.ERA5_q_interp.csh` in the same directory

The NEMO surface boundary fields are 10 m wind, 2 m air temperature, sea-level pressure, snowfall, downward shortwave radiation, downward longwave radiation, precipitation, and specific humidity.

Execution location: The coupled system is configured to launch from the GRIMs scripts. Most execution work is therefore performed under `GRIMs/runs/`.

3.2 Configure the NEMO Namelist

The ocean model, atmospheric model, and coupler must be configured consistently. Begin by editing the NEMO namelist:

```
vi nemo-4.2.2/tests/test_CPL_OASIS/EXP00/namelist_cfg
```

3.2.1 Run-Control Parameters

Parameter	Purpose	Required setting or interpretation
cn_exp	Experiment name	Used as part of NEMO output filenames
nn_itend	Last model time step	Compute from the intended duration and NEMO time step. Example: 21,600 steps x 600 s = 12,960,000 s = 150 days
nn_date0	Model start date	Set the coupled-model start date
ln_rstart	Restart switch	Use `.true.` when a NEMOVAR restart is supplied; use `.false.` when starting from a reanalysis-derived initial field
cn_ocerst_out	Restart-output stem	Set the filename used for generated restart files


```

!-----
! namrun      ! parameters of the run
!-----
  cn_exp      = "eORCA025" ! experience name
  nn_it000    = 1          ! first time step
  nn_itend    = 21600     ! last time step (std 5475)
  nn_date0    = 20121201
  nn_write    = 1440
  ln_rstart   = .false.   ! start from rest (F) or from a restart file (T)
  nn_rstctl   = 0         ! restart control ==> activated only if ln_rstart=T
  !           ! = 0 nn_date0 read in namelist ; nn_it000 : read in namelist
  !           ! = 1 nn_date0 read in namelist ; nn_it000 : check consistency between namelist and restart
  !           ! = 2 nn_date0 read in restart ; nn_it000 : check consistency between namelist and restart
  cn_ocerst_in  = "restart_rebuild" ! suffix of ocean restart name (input)
  cn_ocerst_in  = "restart_DM1_4.2.2_new" ! suffix of ocean restart name (input)
  cn_ocerst_indir = "/rhome/soo/NEMO/input/kisti/eORCA025/nemo_v4.2.2/"
  cn_ocerst_indir = "/rhome/soo/NEMO/couple/restart_check/rebuild_restart_nemo_file/"
  cn_ocerst_out  = "restart" ! suffix of ocean restart name (output)
  cn_ocerst_outdir = "." ! directory in which to write output ocean restarts
  nn_stack      = 14400 ! used only if ln_rst_list = F: output restart frequency (modulo referenced to 1)
/

```

Figure 14. NEMO run-control parameters in `'namelist cfa'`.

3.2.2 Domain and Temperature/Salinity Input

Parameter	Purpose	Setting
ln_read_cfg	Read domain information from a file	Set to <code>`true`</code>
cn_domcfg	Domain-configuration file	Specify the full path, including the filename
ln_tsd_init	Use temperature and salinity initial data	Set to <code>`true`</code>
cn_dir	Initial-data directory	Specify the directory containing the ocean initial fields
sn_tem: filename	Temperature-input filename stem	For <code>`interp_out_y2012m12d01.nc`</code> , enter the stem preceding the date. The final filename must follow <code>`_YYYYmMMdDD.nc`</code>
sn_tem: frequency	Input interval	Enter the interval in hours; use 24 for daily data
sn_tem: variable	Temperature variable	Enter the ocean-temperature variable name in the input file
sn_tem: yearly/monthly	Temporal organization	Set according to the input dataset

```

! namcfg      ! parameters of the configuration                                (default: use namusr_def in namelist_cfg)
!
ln_read_cfg = .true.      ! (=T) read the domain configuration file
!                          ! (=F) user defined configuration                (F => create/check namusr_def)
cn_domcfg = "/rhome/sso/NEMO/input/Kisti/eORCA025/nemo_v4.2.2/test/domain_cfg_4.2.2.nc" ! domain configuration filename
ln_closea = .false.      ! F => suppress closed seas (defined by closea_mask field)
!                          ! from the bathymetry at runtime.

```

Figure 15. NEMO domain-file settings.

```

!-----
&namtsd      !      Temperature & Salinity Data   (init/dmp)                (default: OFF)
!-----
!
!      ! =T   read T-S fields for:
ln_tsd_init = .ture.      ! ocean initialisation
ln_tsd_dmp  = .false.     ! T-S restoring   (see namtra_dmp)

cn_dir = '/rhome/soo/NEMO/input/make_input/make_ocean_init/out/init/'      ! root directory for the T-S data location
!-----
!      !-----
!      ! file name      ! frequency (hours) ! variable ! time interp. ! clim ! 'yearly' ! weights fil
!      !      !      ! (if <0 months) ! name      ! (logical) ! (T/F) ! 'monthly' !
sn_tem = 'interp_out', 24, 'thetao_oras', .false., .false., 'daily', ' ', ' ', ' ', ' ', ' '
sn_sal = 'interp_out', 24, 'so_oras', .false., .false., 'daily', ' ', ' ', ' ', ' ', ' '

```

Figure 16. NEMO temperature and salinity initial-data settings.

3.2.3 Atmospheric Forcing

Set ``cn_dir`` to the directory containing the preprocessed forcing files. Configure ``sn_wndi`` for the zonal wind component and repeat the same pattern for the remaining forcing variables.

Field	Meaning or setting
filename	For `interp_ERA5_u10_y2012m12d01.nc`, enter the stem preceding the date. The filename must end in `_YYYYmMMdDD.nc`.
frequency	Input interval in hours.
variable	Variable name to read from the input file.
yearly/monthly	Temporal organization of the input dataset.
rotation pairing	Used only for wind: `Uwnd` for zonal wind and `Vwnd` for meridional wind.

Apply the same configuration pattern to the remaining fields:

NEMO field	Input variable
wndj	10 m meridional wind
qsr	Downward shortwave radiation
qlw	Downward longwave radiation
tair	2 m air temperature
humi	Specific humidity
prec	Precipitation
snow	Snowfall
slp	Sea-level pressure

```

|-----|
&namsbc_blk      !  namsbc_blk  generic Bulk formula                                (ln_blk =T)
|-----|
|
!          !  bulk algorithm :
ln_NCAR      = .true.      ! "NCAR"      algorithm      (Large and Yeager 2008)

cn_dir = '/rhome/soo/NEMO/input/make_input/make_ocean_init/out/sbc/' !  root directory for the bulk data location
|-----|
|
!          !  file name          ! frequency (hours) ! variable ! time interp. ! clim ! 'yearly' / !
land/sea mask !
!          !          ! (if <0 months) ! name ! (logical) ! (T/F) ! 'monthly' !
filename !
sn_wndi   = 'interp_ERA5_u10', 24, 'u10' , .false. , .false. , 'daily' , '' , 'Uwnd' , ''
sn_wndj   = 'interp_ERA5_v10', 24, 'v10' , .false. , .false. , 'daily' , '' , 'Vwnd' , ''
sn_qsr    = 'interp_ERA5_ssr', 24, 'ssr' , .false. , .false. , 'daily' , '' , '' , ''
sn_qlw    = 'interp_ERA5_strd', 24, 'strd' , .false. , .false. , 'daily' , '' , '' , ''
sn_tair   = 'interp_ERA5_t2m', 24, 't2m' , .false. , .false. , 'daily' , '' , '' , ''
sn_humi   = 'interp_ERA5_q', 24, 'q' , .false. , .false. , 'daily' , '' , '' , ''
sn_prec   = 'interp_ERA5_tp', 24, 'tp' , .false. , .false. , 'daily' , '' , '' , ''
sn_snow   = 'interp_ERA5_snow', 24, 'sf' , .false. , .false. , 'daily' , '' , '' , ''
sn_slp    = 'interp_ERA5_msl', 24, 'msl' , .false. , .false. , 'daily' , '' , '' , ''

```

Figure 17. Example NEMO bulk-forcing entries in `'namelist_cfg'`.

3.3 Configure the OASIS3-MCT Coupler

```
vi runs_gmp/data_oasis3_mct/v4.2.2/namcouple
```

The ``namcouple`` file resides under the GRIMs run directory and is linked into the execution directory at run time. Other fields are already defined for the coupled-model configuration and can normally be used without modification.

1. **RUNTIME:** Specify the total integration time in seconds. The source example uses 12,960,000 s, equivalent to 150 days.
2. **Consistency requirement:** The OASIS3-MCT `RUNTIME` must match the integration duration configured for NEMO.

```

$NFIELDS
15
#####
$RUNTIME
12960000
#####
$NLOGPRT
1 0
# $NNOREST
#Ture
#####
$STRINGS
O_SSTSST o2asst 1 3600 1 rst.nc EXPOUT
1440 1206 384 192 ocen atmo LAG=0
P 0 P 0
MAPPING
weights_eorca025_t126.nc
#
O_TepIce o2atice 1 3600 1 rst.nc EXPOUT
1440 1206 384 192 ocen atmo LAG=0
P 0 P 0
MAPPING
weights_eorca025_t126.nc
#
OIceFrcT o2aicec 1 3600 1 rst.nc EXPOUT
1440 1206 384 192 ocen atmo LAG=0
P 0 P 0
MAPPING
weights_eorca025_t126.nc
#
OSnwTck o2asnwd 1 3600 1 rst.nc EXPOUT
1440 1206 384 192 ocen atmo LAG=0
P 0 P 0
MAPPING
weights_eorca025_t126.nc
#
a2otau_x O_OTaux1 1 3600 1 rst.nc EXPOUT
384 192 1440 1206 atmo ocen LAG=0
P 0 P 0
MAPPING
weights_t126_eorca025.nc

```

Figure 18. Example `namcouple` settings, including `RUNTIME` and field mappings.

3.4 Modify and Recompile the GRIMs Source

```
vi srcs_gmp/src/fcst/phys_main_driver.F90
```

Set the integration duration in the GRIMs source to the same value used for `RUNTIME` in `namcouple`. Recompile the source after making the change.

```
cd srcs_gmp/
make
cd runs_gmp/
```

```

tevp(:, :)=cpl_evp(:, :)
qnso(:, :)=(cpl_dlwsfc(:, :)-cpl_ulwsfc(:, :))-cpl_dtsfc(:, :)-cpl_dqsfc(:, :)
qnsi(:, :)=qnso(:, :)
qsr(:, :)=cpl_sswdn(:, :)-cpl_sswup(:, :)
taux(:, :)=cpl_dusfc(:, :)
tauy(:, :)=cpl_dvsfc(:, :)
ievp(:, :)=cpl_ievp(:, :)
train(:, :)=cpl_raint(:, :)*1.e3
tsnow(:, :)=cpl_snow(:, :)*1.e3

cpl_evp(:, :)= 0.0 ; cpl_raint(:, :)= 0.0
cpl_dlwsfc(:, :)= 0.0 ; cpl_ulwsfc(:, :)= 0.0
cpl_dtsfc(:, :)= 0.0 ; cpl_dqsfc(:, :)= 0.0
cpl_sswdn(:, :)= 0.0 ; cpl_sswup(:, :)= 0.0
cpl_dusfc(:, :)= 0.0 ; cpl_dvsfc(:, :)= 0.0
cpl_ievp(:, :)= 0.0 ; cpl_snow(:, :)= 0.0

!cpl_send(input, intsec, var_id)
! if ( intsec_send .ne. 31968000. ) then
! if ( intsec_send .ne. 8640000. ) then
! if ( intsec_send .ne. 12960000. ) then
!   if (cplmodel .eq. 'ice') then
!     call cpl_send(taux, intsec_send, 5)
!     call cpl_send(tauy, intsec_send, 6)
!     call cpl_send(taux, intsec_send, 7)
!     call cpl_send(tauy, intsec_send, 8)
!     call cpl_send(qsr, intsec_send, 9)
!     call cpl_send(qnso, intsec_send, 10)
!     call cpl_send(qnsi, intsec_send, 11)
!     call cpl_send(tevp, intsec_send, 12)
!     call cpl_send(ievp, intsec_send, 13)
!     call cpl_send(train, intsec_send, 14)
!     call cpl_send(tsnow, intsec_send, 15)
!   else
!     call cpl_send(taux, intsec_send, 2)
!     call cpl_send(tauy, intsec_send, 3)
!     call cpl_send(qsr, intsec_send, 4)
!     call cpl_send(qnso, intsec_send, 5)
!     call cpl_send(tevp, intsec_send, 6)
!   endif
! endif
! write(*,*) "grims_cpl_snd 1", minval(taux), maxval(taux)
! write(*,*) "grims_cpl_snd 2", minval(tauy), maxval(tauy)
! write(*,*) "grims_cpl_snd 3", minval(qsr), maxval(qsr)
! write(*,*) "grims_cpl_snd 4", minval(qnso), maxval(qnso)
! write(*,*) "grims_cpl_snd 5", minval(qnsi), maxval(qnsi)
! write(*,*) "grims_cpl_snd 6", minval(tevp), maxval(tevp)
! write(*,*) "grims_cpl_snd 7", minval(ievp), maxval(ievp)
! write(*,*) "grims_cpl_snd 8", minval(train), maxval(train)
! write(*,*) "grims_cpl_snd 9", minval(tsnow), maxval(tsnow)

```

Figure 19. GRIMs source location where the coupled integration duration is set.

3.5 Configure GRIMs CPU Allocation

```
vi runs_gmp/configure
```

The remaining settings are fixed for the coupled-model configuration. Edit only the CPU allocation assigned to GRIMs in this step.

Variable	Meaning	Source guidance
NCPUS	Threads per MPI task	Fixed at 1
NPES	Number of CPUs assigned to GRIMs	Set to the required GRIMs CPU count
NCPU_PER_NODE	CPUs per compute node	Set for the target system
NCOL	Number of zonal column groups	Use 4 for at least 126 CPUs; otherwise use 2

```

SRCS_DIR=`pwd`/../../srcs_gmp
OUTPUT_DIR=`pwd`
POST_DIR=`pwd`/post
#
# set the environmental variables
#
model=`grep '^MODEL=' $SRCS_DIR/configure-src | \
sed 's/ *$//g' | cut -d '=' -f2`
. ./env/env.s
#
# Need to select resolutions
#
WAVE=126      # define wave number = 62,126,170,214,254,320,426,510,1020,2046
MODEL_LAYER=28 # define number of vertical layer = 18, 28, 42, 64, 128
POST_IGRD=360  # define number of x-axis grids of pressure output file
POST_JGRD=181  # define number of y-axis grids of pressure output file
               # POST_IGRD = 144(2.5deg), 360(1deg), 720(0.5deg), 1440(0.25deg)
               # POST_JGRD = 73(2.5deg), 181(1deg), 361(0.5deg), 721(0.25deg)
POST_LAYER=26  # define number of pressure output file
               # = 12 (rmp), 17(ra2), 21(ra2+4), 26(fn1), 30(fn1+4)
#
# Need to select rmp map projection data.. see ../rmp_domain for more options
#
RMP_MAP=../rmp_domain/rmp_108x85_60km_nps.korea
#
if [ $model = rmp -o $model = rop ]; then
    IGRD=108     # rmp grid selection (IGRD/JGRD)
    JGRD=85      # igrd: product of integer powers 2,3,5; jgrd: odd number
                 # = 54/55, 81/67, 81/87, 108/69, 108/85, 128/85, 162/105
    CIGRD=0      # define number of coarse regional x-axis grids (only for C2R)
    CJGRD=0      # define number of coarse regional y-axis grids (only for C2R)
    IRELX=10     # define relaxation factor in implicit diffusion (only for rmp)
fi
#
# input surface & spherical wave file resolution
#
# T42 (128,64), T62 (192,94), T126 (384,192), T170 (512,256),
# T214 (648,324), T254 (768,384), T320 (968,484), T426 (1280,640),
# T510 (1536,768), T1020 (3072,1536), T2046 (6144,3072)
#
ILONF=384      # for SFC file
ILATG=192      # for SFC file
IJCAP=126      # for SIG file
ILEVS=28       # for SIG file

```

Figure 20. Example GRIMs `runs\_gmp/configure` settings.

```

#
# For DFS, NCOL should be less than MODEL_LAYER,
# more than 2*NPES/JCAP.
#
NCPUS=1         # number of threads (when MARCH = thread or hybrid)
NPES=24         # number of MPI pes (when MARCH = mpi or hybrid)
NCPU_PER_NODE=36 # number of total cores per node
NCOL=2          # number of column (when MARCH=mpi or hybrid)

```

Figure 21. CPU-allocation parameters highlighted in the configure file.

```
./configure gmp
```

3.6 Configure the GRIMs Run Script

```
vi runs_gmp/gmp
```

Edit the script for the selected experiment. The GRIMs start time and integration duration must match the corresponding NEMO settings.

Variable	Description	Setting
RUNNAME	Directory name used to collect output	Specify the experiment/run name
SDATE	Model start time	Set to the coupled-model start time
ENDHOUR	Integration length in hours	For example, '24 * 2' denotes two 24-hour days
INTSIG, INTSFC, INTFLX	Output intervals	Hours; the source example uses 6
INTRESTART	Restart-file interval	Hours; the source example uses 240
FCST_RESTART	Restart switch	Set to 'yes' when restarting
EXECDIR	Output/execution directory	Specify the storage path
INITIAL_SIGMA, INITIAL_SFC	Sigma and surface initial files	Paths to the files generated by P2SIG
SST_ANL_T0	SST input	Path to the ORAS5 SST data
licegrb0, isnogrb0, isstgrb0, ism2grb0	Use of external surface files	Keep at 0 to use the model guess, as in the source setup

3.7 Configure Launch Scripts and Submit the Run

```
vi runs_gmp/runscr/fcst
```

Set the number of CPUs assigned to NEMO in the command that launches `./nemo`. Do not alter the GRIMs CPU count here because it was defined during the configure step. The redirection `1>fcstout.ft\$fh 2>&1` writes standard output and standard error to a log that can be monitored while the model runs.

```
./force_mount.sh
ln -fs $PROG.parm fort.95
/opt/mpich/myrinet/pgi/bin/mpirun -nolocal -np 24 -machinefile $here_dir/hostfile $here_dir/$PROG.x 1
1
else
ln -fs $PROG.parm fort.95
#soojin_couple
# mpirun -machinefile $PBS_NODEFILE -np 24 ./ $PROG.x 1>fcstout.ft$fh 2>&1
mpirun -machinefile $PBS_NODEFILE -bootstrap ssh -np 24 ./ $PROG.x : -np 48 ./nemo 1>fcstout.ft$fh 2>&1
fi
#
stat=$?
if [ $stat -ne 0 ] ; then
echo "`date` $0: Execution failed." >>ERROR.out
exit 8
```

Figure 22. NEMO CPU allocation and log redirection in `runscr/fcst`.

```
vi runs_gmp/runscr/pbs_mpi.head
```

Set the total requested CPUs to the sum of the GRIMs and NEMO allocations. Configure `nodes` as the number of nodes and `ppn` as the number of CPUs per node. On the Air Force No. 3 system, edit `pbspro\_mpi.head` instead.

```
#!/bin/sh
#PBS -l nodes=2:ppn=36
#PBS -V
##PBS -j eo
```

Figure 23. Example PBS node and processor-per-node request.

After completing all settings, launch the coupled model from `runs\_gmp`:

```
cd runs_gmp/
./gmp >& log &
```

3.8 Monitor the Run and Check Outputs

```
cd runs_dir/
ls -lrt
```

File or pattern	Interpretation
ocean.output	NEMO log; inspect this file when an error originates from NEMO.
time.step	NEMO time-step progress log. During execution it shows the current step; after completion it should contain the final step configured in the namelist.
fcst.o or fcstout.ft00	GRIMs standard-output log. If <code>`1>fcstout.ft\$fh 2>&1`</code> is set in <code>`runscr/fcst`</code> , the file is created as <code>`fcstout.ft00`</code> . The final log reports elapsed execution time in seconds.
fcst.e	Error log populated when the model terminates abnormally.
eORCA025_00014400_restart...nc	NEMO restart files.
eORCA025_10d_...grid_T...nc	NEMO model output.
fgb.ft / pgb.ft	GRIMs model output.

```
reset fhour= 2400.000000000000
Wall-clock-time is 92532.3003687633
```

Figure 24. Example GRIMs completion timing reported in the log.